

EV Range & Battery Degradation Modeling for Indian Road, Traffic & Climate Conditions

A focused, portfolio-scale ML project — implementation plan, tech stack, and prerequisites

Scope note: This document covers a deliberately narrowed, buildable version of a broader "Indian EV Mobility Intelligence Platform" idea — one vehicle category, two modeling problems, and a hybrid synthetic + public-data pipeline. It is designed as a CV / portfolio project, not a production platform or startup pitch.

1. Project Overview

Goal: Build an ML pipeline that predicts EV range and models battery degradation for a single Indian vehicle category, explicitly accounting for India-specific conditions — road roughness, stop-start traffic, extreme temperature, and charging behavior — and prove, via an ablation study, that modeling these factors measurably improves accuracy over a generic baseline.

Why this angle: Most public EV range/SoH research and commercial tools are built on Western or Chinese driving cycles (smooth roads, stable climate, consistent traffic). India's conditions — potholes, extreme heat, unreliable slow charging, mixed-load usage — genuinely produce different energy consumption and degradation patterns. This is a real, defensible gap, not just a labeling exercise.

Vehicle category (pick one): Electric 2-wheeler or e-rickshaw. Do not model multiple categories — depth beats breadth for a portfolio project.

2. Scope — What's In, What's Out

In scope

- One vehicle category (2-wheeler or e-rickshaw)
- Two modeling problems: (1) range prediction, (2) battery SoH / degradation trend
- Hybrid data: physics-informed simulator + real public datasets, clearly labeled as such
- Feature engineering for India-specific effects: driving aggressiveness, road roughness proxy, thermal stress index, charging behavior
- An ablation study comparing baseline vs. India-specific-feature models (the headline result)
- A small interactive dashboard to demo predictions
- A written report documenting assumptions, data sources, and limitations

Out of scope

- All six vehicle categories (2W, e-rickshaw, cars, buses, LCVs, commercial EVs) — unbuildable at portfolio scale and dilutes depth

- Real BMS / CAN-bus or real fleet telemetry access — not claiming data you don't have
- A full multi-tenant "platform" product, fleet management UI, or charging-infra optimization layer
- Competing with funded commercial players (Datakrew, 82Volt, Vecmocon, Exicom) — this is a skill demonstration, not a startup
- Exhaustively modeling every factor on the original wishlist at once — select the subset that supports the ablation story

3. Data Strategy

No real fleet telemetry is available for this project, so data comes from a transparent hybrid of simulation and public sources. This must be stated explicitly in the README and report — framed as "synthetic, physics-grounded" data, not implied real-world telemetry.

Data need	Source	Notes
Vehicle energy / driving-cycle data	Physics-informed simulator (equivalent-circuit battery model + driving-cycle generator)	Injects Indian-specific noise: stop-start segments, pothole current spikes, load variation
Real battery degradation curves	Public li-ion aging datasets (e.g. NASA Battery Data Set, or similar open cycling datasets)	Provides genuine capacity-fade ground truth to anchor the degradation model
Weather (temperature, humidity)	IMD data / Open-Meteo API	Real regional and seasonal data — no simulation needed here
Road quality / roughness proxy	OpenStreetMap surface tags, or public pothole datasets from city open-data portals (e.g. Bengaluru, Pune)	Sparse and city-limited — treat as a proxy feature, not ground truth
Charging behavior	Simulated SOC start/end, frequency, fast vs. slow charging sessions	Modeled from assumptions since no real charger logs are available

Optional credibility boost: validate the simulator against even one real vehicle's trip log (e.g. a personal EV or a friend's) to show the synthetic data behaves plausibly.

4. Feature Engineering

Feature	What it captures
Driving aggressiveness score	Frequency/intensity of acceleration and braking events
Road roughness proxy	Pothole/speed-breaker density and surface quality along the route
Thermal stress index	Combined effect of ambient temperature and battery operating temperature
Traffic/stop-start factor	Idling time, average speed, stop frequency
Load factor	Passenger/cargo weight relative to rated capacity
Charging behavior features	Start/end SOC, charge frequency, fast vs. slow charge ratio

5. Models & the Ablation Study

Model 1 — Range prediction: regression model correcting a naive "rated range" baseline using trip-condition features.

Model 2 — Degradation / SoH trend: time-series model estimating capacity fade or remaining SoH over charge cycles, under different simulated Indian conditions.

The ablation study is the headline deliverable — it proves the India-specific features matter, rather than just presenting a model that works:

Model variant	Features used	Result (illustrative)
Baseline	Rated range only	Highest error
+ Weather	Rated range + temperature/humidity	Lower error
+ Road / Traffic	+ roughness, stop-start factor	Lower still
Full model	All India-specific features	Lowest error

This table/chart is the single most important output — it is what goes on the CV, in the portfolio write-up, and is the anchor of any interview conversation about this project.

6. Tech Stack

Layer	Tools
Language / core	Python 3.10+
Data handling	pandas, numpy
Battery / physics simulation	Custom equivalent-circuit model (numpy/scipy), optionally PyBaMM for a more rigorous electrochemical model

Layer	Tools
Modeling	scikit-learn (baseline regressors), XGBoost/LightGBM (tabular models), statsmodels or a simple RNN/LSTM (PyTorch) for degradation time-series
Feature engineering / geo	geopandas, OSMnx (OpenStreetMap data), requests (weather APIs)
Visualization	matplotlib, seaborn, plotly
Dashboard	Streamlit (fastest path) or a small React app
Experiment tracking (optional)	MLflow or simple CSV/JSON logging for ablation results
Version control	Git + GitHub, with a clearly written README
Environment	venv or conda; requirements.txt or pyproject.toml

7. Prerequisites

Knowledge / skills

- Python and pandas/numpy fluency
- Core ML: regression, evaluation metrics (MAE/RMSE), train/test/validation splitting
- Basic time-series concepts for the degradation model
- Basic understanding of battery behavior: SoC, SoH, C-rate, internal resistance (a few hours of reading on equivalent-circuit battery models is enough — this doesn't require an electrochemistry background)
- Comfort working with APIs (weather data) and geospatial data (OSM) — new for many, but well-documented and learnable within the project timeline
- Git basics for version control and a clean public repo

Accounts / access (all free)

- GitHub account for the repo
- Open-Meteo API (no key required) or IMD public data access for weather
- Access to a public li-ion battery aging dataset (e.g. NASA Battery Data Set — free download)
- OpenStreetMap / Overpass API access for road tags (free, no key required)
- Optional: a city open-data portal account if using a specific pothole dataset (e.g. Bengaluru/Pune)

Hardware

- A standard laptop is sufficient — no GPU required for the tabular models
- A GPU (or free-tier Colab/Kaggle notebook) is helpful only if using an LSTM/RNN for the degradation model, but not mandatory

8. Implementation Plan (4–6 Weeks)

Week	Focus	Key tasks
Week 1	Setup & data sourcing	Set up repo/environment; download battery aging dataset; wire up weather API; gather road-quality proxy data for chosen city/region; finalize vehicle category
Week 2	Simulator build	Build equivalent-circuit battery model; build driving-cycle generator with Indian-condition noise (stop-start, pothole spikes, load variation); sanity-check outputs
Week 3	Feature engineering	Build driving aggressiveness score, road roughness proxy, thermal stress index, charging behavior features; merge simulated + real data into a unified dataset
Week 4	Modeling	Train baseline and full range-prediction models; train degradation/SoH model; run initial evaluation (MAE/RMSE)
Week 5	Ablation study & visualization	Run staged ablation (baseline -> +weather -> +road/traffic -> full model); produce error-comparison charts and feature-importance plot

Week	Focus	Key tasks
Week 6	Dashboard, report & polish	Build Streamlit dashboard; write the methodology/limitations report; finalize README; prepare the CV write-up and repo polish

Risk note: the simulator (Week 2) is the hardest and most time-sensitive step. If it stalls, de-scope by simplifying the battery model or leaning more heavily on the public aging dataset for degradation ground truth, using the simulator mainly for driving-condition variation.

9. Final Output Stack

- A structured, documented GitHub repo (data/simulator/features/models/notebooks/dashboard/report)
- Two trained models: range predictor and degradation/SoH model
- An ablation table/chart showing % error reduction from India-specific features (headline result)
- Visualizations: error curves, degradation curves, feature-importance plot
- A working demo dashboard (Streamlit or small web app)
- A 1-2 page written report covering methodology, data sources, and limitations
- A one-line CV bullet + link to the repo/dashboard

10. Suggested CV Write-Up

EV Range & Battery Degradation Modeling for Indian Conditions — Built an ML pipeline combining physics-based battery simulation, public battery-aging datasets, and Indian weather/road data to predict range and SoH degradation for electric 2-wheelers; demonstrated a measurable error reduction from modeling India-specific driving, road, and climate features versus a baseline model.

Keep the framing honest: describe the data as "synthetic, physics-grounded, and fused with public datasets" — never imply real fleet or BMS telemetry access that wasn't actually used.